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BRANCH AND SECTION: CSM-B

COURSE NAME: FULL STACK WEB DEVELOPMENT

COURSE CODE: 23CM4121

Experiment: 04

Date:

Title: Build an Interactive Flashcard App using HTML, CSS and JavaScript

Objective:

To create an interactive flashcard web application using HTML for structure, embedded CSS for styling, and JavaScript to handle click events, DOM manipulation, and dynamic content updates.

System Requirements:

- Hardware: Processor (e.g., Core i5), RAM (e.g., 8GB), Storage.
- Software: Operating System (e.g., Windows 11 / Linux), Code Editor (e.g., VS Code), Browser (e.g., Chrome), Runtime/Database (e.g., Node.js, MongoDB).

Source Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Beginner Flashcard App</title>

  <style>
    /* 1. Overall Page Layout */
    body {
      font-family: Arial, sans-serif;
      background-color: #eef2f3;
      display: flex;
      justify-content: center;
      align-items: center;
      min-height: 100vh;
      margin: 0;
    }

    .container {
      text-align: center;
      width: 320px;
    }

    /* 2. Flashcard Styling */
    .card {
      background-color: white;
```

```

border-radius: 12px;
padding: 40px 20px;
box-shadow: 0 4px 15px rgba(0, 0, 0, 0.1);
cursor: pointer;
min-height: 120px;
display: flex;
flex-direction: column;
justify-content: center;
align-items: center;
transition: transform 0.2s ease, background-color 0.3s ease;
}

.card:hover {
  transform: translateY(-3px);
}

.card-title {
  font-size: 0.85rem;
  color: #888;
  text-transform: uppercase;
  letter-spacing: 1px;
  margin-bottom: 10px;
}

.card-text {
  font-size: 1.25rem;
  color: #333;
  font-weight: bold;
  margin: 0;
}

/* Active state when answer is showing */
.card.revealed {
  background-color: #e8f5e9;
}

.hint-text {
  font-size: 0.8rem;
  color: #aaa;
  margin-top: 15px;
}

/* 3. Button Styling */
.btn {
  margin-top: 20px;
  padding: 12px 24px;
  font-size: 1rem;
  background-color: #3498db;
  color: white;
  border: none;
  border-radius: 6px;
  cursor: pointer;
  width: 100%;
}

```

```

        .btn:hover {
            background-color: #2980b9;
        }
    </style>
</head>
<body>

    <div class="container">
        <!-- The Flashcard -->
        <div class="card" id="flashcard">
            <span class="card-title" id="cardLabel">Question</span>
            <p class="card-text" id="cardText">Loading question...</p>
            <span class="hint-text">(Click card to flip)</span>
        </div>

        <!-- Next Question Button -->
        <button class="btn" id="nextBtn">Next Question</button>
    </div>

    <!-- Linking the JavaScript File -->
    <script src="script.js"></script>

</body>
</html>

```

script.js

```

// Step 1: Store our data (Array of question/answer objects)
const flashcards = [
    { question: "What does HTML stand for?", answer: "HyperText Markup Language" },
    { question: "What language styles a webpage?", answer: "CSS (Cascading Style Sheets)" },
    { question: "Which HTML tag connects a JS file?", answer: "<script src='...'>" },
    { question: "What tool does JavaScript use to listen for clicks?", answer: "addEventListener()" }
];

// Track current question index & state
let currentIndex = 0;
let isShowingAnswer = false;

// Step 2: Select elements from HTML using IDs
const flashcardElement = document.getElementById("flashcard");
const cardLabel = document.getElementById("cardLabel");
const cardText = document.getElementById("cardText");
const nextBtn = document.getElementById("nextBtn");

// Step 3: Function to display current flashcard
function loadCard() {
    isShowingAnswer = false;
    cardLabel.textContent = "Question";
    cardText.textContent = flashcards[currentIndex].question;
    flashcardElement.classList.remove("revealed");
}

```

```
// Step 4: Event Listener - Flip Card on Click
flashcardElement.addEventListener("click", function () {
  if (isShowingAnswer) {
    // Flip back to Question
    cardLabel.textContent = "Question";
    cardText.textContent = flashcards[currentIndex].question;
    flashcardElement.classList.remove("revealed");
    isShowingAnswer = false;
  } else {
    // Show Answer
    cardLabel.textContent = "Answer";
    cardText.textContent = flashcards[currentIndex].answer;
    flashcardElement.classList.add("revealed");
    isShowingAnswer = true;
  }
});

// Step 5: Event Listener - Next Button Click
nextBtn.addEventListener("click", function () {
  // Move to next card index (loop back to start if at the end)
  currentIndex = (currentIndex + 1) % flashcards.length;
  loadCard();
});

// Load the first card when the page opens
loadCard();
```

STEPS TO EXECUTE:

- 1) Open VS Code
- 2) Click on Run in the menu
- 3) Click on Debug to Run
- 4) Select browser and click run

OUTPUT:

